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Inference about the change-point from cumulative sum tests

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Summary

The point of change in mean in a sequence of normal random variables can be estimated from a cumulative sum test scheme. The asymptotic distribution of this estimate and associated test statistics are derived and numerical results given. The relation to likelihood inference is emphasized. Asymptotic results are compared with empirical sequential results, and some practical implications are discussed.

1. Introduction

The cumulative sum scheme for detecting distributional change in a sequence of random variables is a well-known technique in quality control, dating from the paper of Page (1954) to the recent expository account by van Dobben de Bruyn (1968). Throughout the literature on cumulative sum schemes the emphasis is placed on tests of departure from initial conditions. The purpose of this paper is to examine a secondary aspect: estimation of the index τ in a sequence $\{x_t\}$, where the departure from initial conditions has taken place. The work is closely related to an earlier paper by Hinkley (1970), in which maximum likelihood estimation and inference were discussed.

We consider specifically sequences of normal random variables $x_1, \ldots, x_T$, say, where initially the mean θ_0 and the variance σ^2 are known. A cumulative sum, cusum, scheme is used to detect possible change in mean from θ_0, and for simplicity suppose that it is a one-sided scheme for detecting decrease in mean. Then the procedure is to compute the cumulative sums

$$S_0 = 0, \quad S_t = \sum_{j=1}^{t} (x_j - \theta_0 + \delta\sigma) \quad (t = 1, 2, \ldots) \tag{1·1}$$

and to reject the hypothesis that there is no change if

$$S_T - \max_{t<T} S_t < -h, \tag{1·2}$$

where $\delta > 0$ and $h > 0$ are chosen to give the scheme specified properties. If the scheme is sequential, these properties are in terms of average run length and T is a random variable, defined by the stopping rule (1·2). If the scheme is used for a given sample of size T, the properties of the scheme are in terms of test power. In either case it is common to specify a 'rejectable quality level' θ_0^*, and then to determine δ and h so that the scheme has specified average run length or power both when the mean changes to θ_0^* and when no change from θ_0 occurs. The rejectable quality level θ_0^* and δ must be chosen with some care. For it is clear from (1·1) and (1·2) that if the actual change in mean is less than $\delta\sigma$, then the increments $x_j - \theta_0 + \delta\sigma$ always have positive mean and the scheme will be ineffective as a detector of the change. Typically δ is close to $\frac{1}{2}(\theta_0 - \theta_0^*)/\sigma$, so that θ_0^* must be sensibly chosen.

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The usual estimate of the point of change in mean is the index of the maximum cusum, which we shall denote by

$$\tilde{\tau} = \inf(t: S_t \geqslant S_u, u = 1, \ldots, T). \qquad (1\cdot 3)$$

We shall concentrate in what follows on the distribution of $\tilde{\tau}$, with two aims in view. First, to illustrate the variation in $\tilde{\tau}$ for particular values of actual mean change and cusum parameter δ. Secondly, to compare $\tilde{\tau}$ with the maximum likelihood estimate $\hat{\tau}$. The reason for this comparison is that extra work is involved in getting the maximum likelihood estimate, and it would be helpful to know how much information is lost by not doing it. A further point is that both $\tilde{\tau}$ and $\hat{\tau}$ are inefficient inferential statistics. An analogue of the likelihood ratio test statistic exists for the cusum scheme, which is simpler to use than the likelihood ratio test statistic itself. We discuss the asymptotic distributions of $\tilde{\tau}$, $\hat{\tau}$ and the likelihood ratio test statistics in § 2 under the assumption that τ and $T-\tau$ are large. Then in § 3 we examine the accuracy of these distributions as finite sample approximations, with particular emphasis on the sequential cusum scheme where T is variable.

We shall restrict ourselves to the one-sided scheme defined by (1·1) and (1·2) for simplicity. The results for the other one-sided scheme and the two-sided scheme can be inferred with little difficulty. For example, the two-sided scheme is made up from two one-sided schemes, only one of which gives a rejection; this is the one used to estimate the change-point.

2. Asymptotic theory, fixed T

Suppose that we observe a sequence $x_1, \ldots, x_T$ such that

$$x_i = \begin{cases} \theta_0 + e_i & (i = 1, \ldots, \tau), \\ \theta_1 + e_i & (i = \tau+1, \ldots, T), \end{cases} \qquad (2\cdot 1)$$

where the errors $e_1, \ldots, e_T$ are independent $N(0, \sigma^2)$ variables with σ^2 known. We shall assume throughout this section that T is fixed, not determined by (1·2), since in the theoretical analysis it is convenient to have $T \gg \tau$. The cusum test scheme is defined by (1·1) and (1·2) for some choice of δ and h, and $\tilde{\tau}$ as defined in (1·3) depends on δ.

We assume θ_0 is known, but that θ_1 is not. However, we do assume $\theta_1 < \theta_0$, since we are only interested in estimating τ if the cusum test implies its existence. Further, it is necessary to assume that $\theta_0 - \theta_1 > \delta\sigma$ in order to ensure nondegenerate theory; see the remarks in § 1. The asymptotic theory developed in this section will assume both τ and $T-\tau$ to be infinitely large.

2·1. *Relation between $\tilde{\tau}$ and $\hat{\tau}$*

It is easy to see from (2·1) that the maximum likelihood estimate $\hat{\tau}$ is the value of t which maximizes

$$Z_t^2 = (T-t)^{-1}\left\{\sum_{i=t+1}^{T}(x_i - \theta_0)\right\}^2 \quad (t = 1, \ldots, T-1). \qquad (2\cdot 2)$$

When $\theta_1 < \theta_0$, $\hat{\tau}$ is asymptotically equivalent to the value of t which maximizes $\{-Z_t\}$. In fact $\{-Z_t\}$ and $\{S_t\}$ are equivalent, since by (1·1) and (2·2)

$$Z_t = (T-t)^{-\frac{1}{2}}\{S_T - S_t - (T-t)\,\delta\sigma\}. \qquad (2\cdot 3)$$

Incidentally this relationship shows that $\hat{\tau}$ can be obtained directly from a cusum plot by superimposing the parabolae

$$y = S_T + c(T-t)^{\frac{1}{2}} - (T-t)\,\delta\sigma \quad (c > 0; t = T, T-1, \ldots, 1)$$

on the plot with vertex at (T, S_T) and increasing c until one cusum lies above the curve, namely $S_{\hat{\tau}}$. This procedure is directly related to Barnard's (1959) parabolic mask cusum schemes. In practice this graphica method of computing $\hat{\tau}$ would be difficult to apply.

It can be seen from (2·3) that $\tilde{\tau}$ will be asymptotically biased, because $\hat{\tau}$ is asymptotically unbiased; moreover, $\tilde{\tau}$ is less efficient than $\hat{\tau}$. This can be seen another way by observing that when θ_1 is known the maximum likelihood estimate is simply the cusum estimate using $\delta = \frac{1}{2}(\theta_0 - \theta_1)/\sigma$ in (1·1). Thus for general δ the cusum scheme is a rotation of the optimum

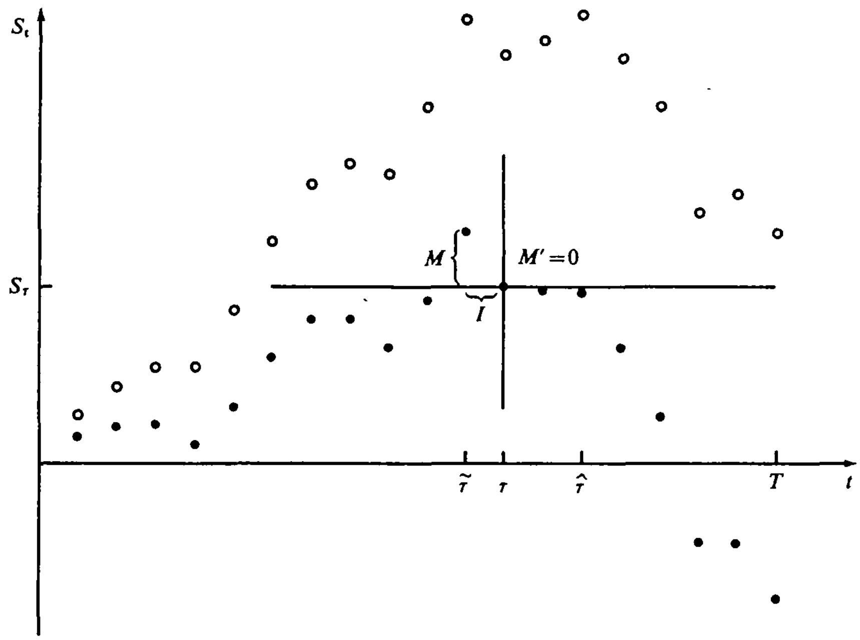

Fig. 1. Cumulative sum plots for $\delta = \Delta$ (likelihood for known θ_1) and $\delta = \frac{1}{2}\Delta$, with random walk structure for $\delta = \frac{1}{2}\Delta$. ● $\delta = \frac{1}{2}\Delta$; ○ $\delta = \Delta$.

scheme about the origin, resulting in a bias in $\tilde{\tau}$. This is illustrated in Fig. 1. That $\tilde{\tau}$ is less efficient than $\hat{\tau}$ follows because sample information about θ_1 is not used in $\tilde{\tau}$. Of course if $\delta \simeq \frac{1}{2}(\theta_0 - \theta_1)/\sigma$ this does not matter, because we have then effectively guessed θ_1. A full discussion of the maximum likelihood estimate is given by Hinkley (1970), who also indicates that the asymptotic distribution of $\hat{\tau}$ is unaffected by not knowing θ_1 and/or θ_0, essentially because these parameters can be consistently estimated. The asymptotic distribution of $\hat{\tau}$ is therefore the special case of that for $\tilde{\tau}$ when $\delta = \frac{1}{2}(\theta_0 - \theta_1)/\sigma$.

2·2. *Asymptotic distribution of $\tilde{\tau}$*

The derivation of the asymptotic distribution of $\tilde{\tau}$ parallels that given by Hinkley (1970) for $\hat{\tau}$. First, we note that the sequence $S_1, \ldots, S_T$ defines two random walks which measure S_t relative to S_τ, that is

$$\mathbf{W} = \{0, S_{\tau-1} - S_\tau, \ldots, S_1 - S_\tau\},$$
$$\mathbf{W}' = \{0, S_{\tau+1} - S_\tau, \ldots, S_T - S_\tau\}.$$

Now assume τ and $T - \tau$ to be infinitely large, and let

$$y_j = -(x_{\tau-j+1} - \theta_0 + \delta\sigma)/\sigma \quad (j = 1, 2, \ldots),$$
$$y'_j = (x_{\tau+j} - \theta_0 + \delta\sigma)/\sigma \quad (j = 1, 2, \ldots), \tag{2·4}$$

so that

$$\mathbf{W} = \left\{0, \sum_{j=1}^{k} y_j \quad (k = 1, 2, \ldots)\right\},$$

$$\mathbf{W}' = \left\{0, \sum_{j=1}^{k} y'_j \quad (k = 1, 2, \ldots)\right\}. \tag{2·5}$$

By (2·4) and (2·1) the increments y_j and y'_j are respectively $N(-\delta, 1)$ and $N(\delta - 2\Delta, 1)$ with $\Delta = \frac{1}{2}(\theta_0 - \theta_1)/\sigma$. Recall that we are assuming $\delta < 2\Delta$.

By the construction of $\mathbf{W}$ and $\mathbf{W}'$ there is an equivalence between $\tilde{\tau} - \tau$ and the index of the larger of the two random walk maxima; if both $\mathbf{W}$ and $\mathbf{W}'$ remain below zero, then S_τ is the largest cusum and $\tilde{\tau} = \tau$. To exploit the random walk representation we make the following definitions. Let

$$U = \max_{k \geqslant 1} \sum_{j=1}^{k} y_j, \quad M = \max(0, U),$$

$$U' = \max_{k \geqslant 1} \sum_{j=1}^{k} y'_j, \quad M' = \max(0, U');$$

$$I = \inf\left(k : M = \sum_{j=1}^{k} y_j\right), \quad \text{with} \quad I = 0 \quad \text{if} \quad M = 0;$$

$$I' = \inf\left(k : M = \sum_{j=1}^{k} y'_j\right), \quad \text{with} \quad I' = 0 \quad \text{if} \quad M' = 0.$$

Then it follows from (1·3) and (2·5) that

$$\tilde{\tau} - \tau = \begin{cases} -n & (I = n, M > M', M > 0), \\ 0 & (M = M' = 0), \\ n & (I' = n, M' > M, M' > 0). \end{cases} \tag{2·6}$$

This random walk structure is illustrated in Fig. 1, where for $\delta = \frac{1}{2}\Delta$ we have $I' = M' = 0$, $I = 1$ and $M > 0$, so that $\tilde{\tau} - \tau = -1$.

Next define the probability measures

$$\beta_n(x; \delta)\, dx = \operatorname{pr}(I = n, x \leqslant M < x + dx; \delta),$$
$$\alpha(x; \delta) = \operatorname{pr}(M \leqslant x; \delta),$$
$$q_n(\delta) = \operatorname{pr}(I = n; \delta).$$

Then it follows from (2·6) and the preceding definitions that the distribution

$$p_n(\delta, \Delta) = \operatorname{pr}(\tilde{\tau} - \tau = n; \delta, \Delta)$$

is given by

$$p_n(\delta, \Delta) = \begin{cases} q_{-n}(\delta) - \int_0^\infty \{1 - \alpha(u; 2\Delta - \delta)\}\beta_{-n}(u; \delta)\, du & (n < 0), \\ \alpha(0; \delta)\,\alpha(0; 2\Delta - \delta) & (n = 0), \\ q_n(2\Delta - \delta) - \int_0^\infty \{1 - \alpha(u; \delta)\}\beta_n(u; 2\Delta - \delta)\, du & (n > 0). \end{cases} \tag{2·7}$$

Unfortunately explicit expressions do not exist for any of the quantities in (2·7), with the exception of $\alpha(0; \lambda)$. However, an explicit expression does exist for the double Laplace transform of $\beta_n(x; \lambda)$, from which we can obtain recurrence relations for $q_n(\lambda)$ and the

Laplace transform of $\beta_n(x;\lambda)$ with respect to x; see Hinkley (1970). It follows, then, that in order to compute the distribution (2·7) we need to express $1-\alpha(x;\lambda)$ as an exponential series. This can be done using numerical solution of the integral equation

$$1-\alpha(x;\lambda) = 1-\Phi(x+\lambda)+\int_0^\infty \{1-\alpha(x;\lambda)\}\,\phi(x-u+\lambda)\,du. \tag{2·8}$$

Here $\Phi(.)$ represents the normal distribution and $\phi(.)$ the normal density function. In fact $1-\alpha(x;\lambda)$ can be accurately approximated by the sum of only two exponential terms. A full account of this approximation and the computation of (2·7) is given by Hinkley (1970). Here it is enough to say that $p_n(\delta,\Delta)$ can be computed quite easily and accurately, the relative error being less than 0·003 for all cases described in this paper.

Table 1. *Asymptotic bias of* $\tilde{\tau}$

δ \ Δ	0·5	0·6	0·7	0·8	0·9	1·0	1·1	1·2	1·3	1·4	1·5
0·3	−4·03	−4·41	−4·49	−4·69	−4·74	−4·78	−4·80	−4·81	−4·81	−4·81	−4·81
0·4	−1·71	−2·30	−2·56	−2·69	−2·76	−2·81	−2·83	−2·84	−2·84	−2·84	−2·84
0·5	0·00	−0·97	−1·35	−1·53	−1·64	−1·69	−1·72	−1·74	−1·75	−1·75	−1·75
0·6	—	0·00	−0·59	−0·86	−0·99	−1·07	−1·11	−1·14	−1·15	−1·15	−1·15
0·7	—	—	0·00	−0·29	−0·58	−0·68	−0·73	−0·77	−0·79	−0·80	−0·80
0·8	—	—	—	0·00	−0·26	−0·40	−0·48	−0·52	−0·55	−0·57	−0·58
0·9	—	—	—	—	0·00	−0·19	−0·20	−0·35	−0·38	−0·40	−0·41
1·0	—	—	—	—	—	0·00	−0·14	−0·21	−0·26	−0·29	−0·30

Table 2. 1 % *and* 99 % *points of asymptotic distribution of* $\tilde{\tau}-\tau$

		99 % points Δ										
δ	1 % point all $\Delta \geqslant \delta$	0·5	0·6	0·7	0·8	0·9	1·0	1·1	1·2	1·3	1·4	1·5
0·3	−46	8	5	4	3	2	2	2	1	1	1	1
0·4	−26	11	6	4	3	2	2	2	2	1	1	1
0·5	−17	17†	9	5	4	3	2	2	2	2	1	1
0·6	−12	—	12†	7	5	3	3	2	2	2	1	1
0·7	−9	—	—	9†	6	4	3	2	2	2	2	1
0·8	−7	—	—	—	7†	5	4	3	2	2	2	2
0·9	−6	—	—	—	—	6†	4	3	3	2	2	2
1·0	−5	—	—	—	—	—	5†	4	3	2	2	2

† Maximum likelihood estimate.

The distribution of $\tilde{\tau}$ is very dispersed if δ or Δ is less than 0·5, and the computation is correspondingly lengthy. Numerical results are therefore restricted here to cases where $\Delta \geqslant 0{\cdot}5$, that is $\theta_0-\theta_1 \geqslant \sigma$. Table 1 gives the bias $E(\tilde{\tau}-\tau)$ for $\Delta = 0{\cdot}5\,(0{\cdot}1)\,1{\cdot}5$ and $\delta = 0{\cdot}3\,(0{\cdot}1)\,1{\cdot}0$. To indicate the spread of the distributions Table 2 gives the 1 % and 99 % points. For each δ, the 1 % point is constant over the values of Δ, so the 1 % point is recorded once in the first column. Results for $\delta > \Delta$ have been omitted from both tables since, by (2·7), $p_n(\delta,\Delta) = p_{-n}(2\Delta-\delta,\Delta)$. As an example of the way the distribution of $\tilde{\tau}$ changes when δ moves away from Δ, Fig. 2 is a plot of the distributions of $\tilde{\tau}-\tau$ for the cases $\delta = 0{\cdot}5$, 0·7 and 1·0 when $\Delta = 1{\cdot}0$.

The rapid change in distribution of $\tilde{\tau}$ as $|\delta-\Delta|$ increases is clearly illustrated in Table 2 and Fig. 2. The bias in $\tilde{\tau}$ is large if $\delta \leqslant 0{\cdot}4$ and $\Delta > 0{\cdot}5$. As a whole the results indicate serious deficiencies in $\tilde{\tau}$ as an estimate compared to $\hat{\tau}$. Note that if δ underestimates Δ, then the change-point will also be underestimated. As $\delta \to 0$, $\mathrm{pr}(\tilde{\tau} > \tau)$ decreases monotonically to zero. A short table of the asympotic distribution of $\hat{\tau}$ is given by Hinkley (1970).

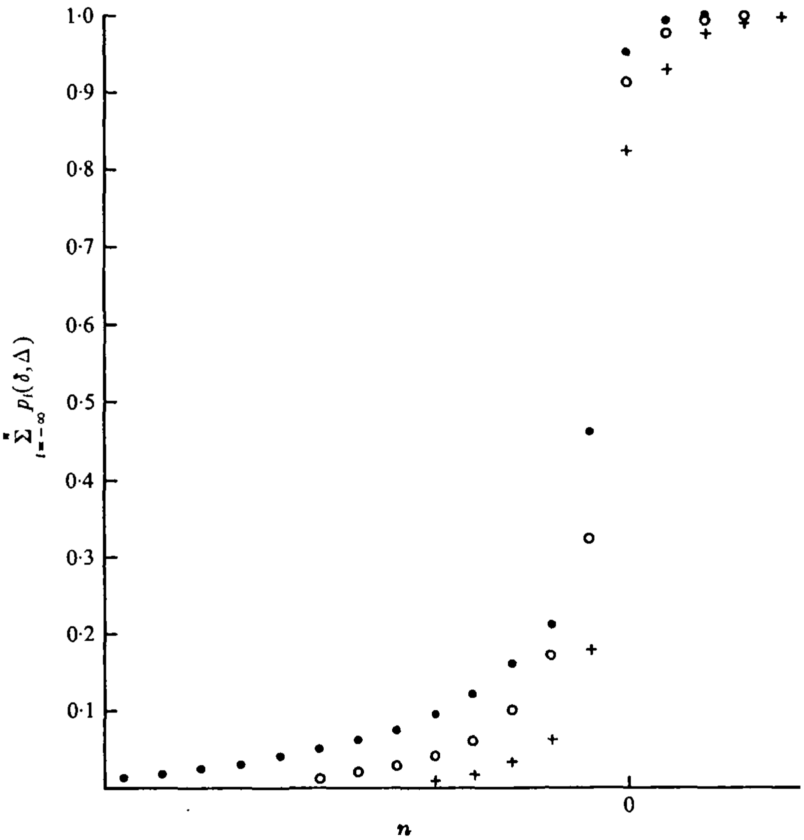

Fig. 2. Asymptotic cumulative distribution functions of $\tilde{\tau} - \tau$ for $\delta = 0{\cdot}5,\ 0{\cdot}7,\ 1{\cdot}0$ and $\Delta = 1{\cdot}0$. ● $\delta = 0{\cdot}5$, ○ $\delta = 0{\cdot}7$, + $\delta = \Delta = 1{\cdot}0$.

2·3. *Inference about* τ

As well as estimating τ from the cusum scheme one would frequently want to construct a confidence interval for τ or test a hypothetical change-point τ_0. The estimates $\tilde{\tau}$ and $\hat{\tau}$ can be used, but both are inefficient by comparison to the likelihood, essentially because no sufficient statistics for τ exist except the observations. Likelihood ratio tests of $H_0: \tau = \tau_0$ were discussed by Hinkley (1970), who derived their null asymptotic properties.

Suppose, for example, that we test H_0 against the two-sided alternative H_2: $\tau \neq \tau_0$. Then the likelihood ratio test statistic, for θ_1 unknown, is

$$\Lambda_2 = \frac{1}{2\sigma^2}(Z_{\hat{\tau}}^2 - Z_{\tau_0}^2)$$

with null asymptotic distribution function

$$P_2(x; \Delta) = [\alpha\{x/(2\Delta); \Delta\}]^2. \tag{2·9}$$

If θ_1 were known, Λ_2 would be replaced by

$$\frac{\theta_0-\theta_1}{\sigma^2}\left[\max_t \sum_{i=1}^{t}\{x_i-\tfrac{1}{2}(\theta_0+\theta_1)\}-\sum_{i=1}^{\tau_0}\{x_i-\tfrac{1}{2}(\theta_0+\theta_1)\}\right],$$

which suggests as a simple alternative to Λ_2 the cusum statistic

$$\Lambda_2^* = \frac{1}{\sigma}(S_{\hat\tau}-S_{\tau_0}).$$

Table 3. *Significance points of* Λ_2^*

$\delta \backslash \Delta$	0·5	0·6	0·7	0·8	0·9	1·0	1·1	1·2	1·3	1·4	1·5
						5 % test					
0·3	4·44	4·44	4·44	4·44	4·44	4·44	4·44	4·44	4·44	4·44	4·44
0·4	3·39	3·22	3·17	3·17	3·17	3·17	3·17	3·17	3·17	3·17	3·17
0·5	3·09	2·65	2·50	2·44	2·42	2·42	2·42	2·42	2·42	2·42	2·42
0·6	—	2·48	2·15	2·02	1·96	1·92	1·92	1·92	1·92	1·92	1·92
0·7	—	—	2·05	1·80	1·68	1·62	1·59	1·57	1·57	1·57	1·57
0·8	—	—	—	1·73	1·52	1·41	1·36	1·33	1·31	1·30	1·30
0·9	—	—	—	—	1·48	1·30	1·20	1·15	1·12	1·09	1·08
1·0	—	—	—	—	—	1·27	1·12	1·03	0·97	0·93	0·90
						2 % test					
0·3	5·94	5·94	5·94	5·94	5·94	5·94	5·94	5·94	5·94	5·94	5·94
0·4	4·46	4·32	4·31	4·31	4·31	4·31	4·31	4·31	4·31	4·31	4·31
0·5	4·02	3·51	3·34	3·33	3·33	3·33	3·33	3·33	3·33	3·33	3·33
0·6	—	3·26	2·87	2·74	2·69	2·68	2·68	2·68	2·68	2·68	2·68
0·7	—	—	2·71	2·41	2·29	2·23	2·22	2·22	2·22	2·22	2·22
0·8	—	—	—	2·30	2·06	1·95	1·88	1·88	1·88	1·88	1·88
0·9	—	—	—	—	1·99	1·79	1·69	1·65	1·61	1·61	1·61
1·0	—	—	—	—	—	1·74	1·57	1·48	1·43	1·40	1·39
						1 % test					
0·3	7·10	7·10	7·10	7·10	7·10	7·10	7·10	7·10	7·10	7·10	7·10
0·4	5·29	5·22	5·18	5·18	5·18	5·18	5·18	5·18	5·18	5·18	5·18
0·5	4·72	4·17	4·04	4·03	4·03	4·03	4·03	4·03	4·03	4·03	4·03
0·6	—	3·84	3·41	3·30	3·27	3·26	3·26	3·26	3·26	3·26	3·26
0·7	—	—	3·21	2·87	2·76	2·73	2·72	2·72	2·72	2·72	2·72
0·8	—	—	—	2·74	2·47	2·37	2·32	2·30	2·30	2·30	2·30
0·9	—	—	—	—	2·37	2·15	2·06	2·02	2·01	2·00	1·99
1·0	—	—	—	—	—	2·08	1·90	1·81	1·77	1·75	1·74

When H_0 is true $\Lambda_2^* = \max(M, M')$ with M and M' as defined in § 2·2. Therefore the asymptotic null distribution of Λ_2^* is

$$P_2^*(x;\delta,\Delta) = \alpha(x;\delta)\,\alpha(x;2\Delta-\delta). \tag{2·10}$$

When $\delta = \Delta$, Λ_2 and Λ_2^* are asymptotically equivalent, apart from the factor 2Δ.

Values of the distribution function $P_2^*(x;\delta,\Delta)$ are relatively easy to compute, using numerical solution of (2·8) for $\alpha(x;\lambda)$. Table 3 gives 95, 98 and 99 % quantiles of Λ_2^* for $0·3 \leqslant \delta \leqslant 1·0$ and $0·5 \leqslant \Delta \leqslant 1·5$. Hinkley (1970) gives the corresponding quantiles for both one-sided and two-sided likelihood ratio test statistics.

One might be interested in a one-sided test of H_0 with alternative $\tau < \tau_0$; see § 4. Then corresponding to Λ_2^* is the one-sided cusum test statistic

$$\Lambda_1^* = \frac{1}{\sigma}\max_{t\leqslant\tau_0}(S_t - S_{\tau_0}), \tag{2·11}$$

whose null distribution is $P_1^*(x;\delta) = \alpha(x;\delta)$, independent of Δ. Note that

$$P_2^*(x;\delta,\Delta) \to P_1^*(x;\delta)$$

as $\Delta \to \infty$, so that the final column in Table 3 gives approximate percentage points for Λ_1^*.

The following relations are useful for interpolation from Table 3. If

$$P_2^*\{l_\alpha(\delta,\Delta);\delta,\Delta\} = 1-\alpha, \tag{2·12}$$

then

(i) $\log\{l_\alpha(\delta,\Delta)-l_\alpha(\delta,\infty)\}$ is approximately linear in δ for $\Delta \geqslant \delta$;
(ii) $\log\{\Delta l_\alpha(\Delta,\Delta)\}$ is very close to linearity in Δ; and
(iii) $\log\{\delta l_\alpha(\delta,\infty)\}$ is very close to linearity in δ,

all for α, δ and Δ in the range of Table 3. These are based on empirical evidence and the asymptotic behaviour of $\alpha(x;\lambda)$ for large x.

Restricting attention to two-sided tests, we have four statistics available for inference about τ: the estimates $\tilde{\tau}$ and $\hat{\tau}$, the likelihood ratio statistic Λ_2 and its cusum analogue Λ_2^*. In order to make a full comparison of these statistics we need to compute the nonnull distribution of Λ_2^*. To do this, note that if $\tau = \tau_0+r$, then

$$\Lambda_2^* = \frac{1}{\sigma}\max\{\max_{t\leqslant\tau-r}(S_t-S_{\tau-r}), \max_{t\geqslant\tau-r}(S_t-S_{\tau-r})\}. \tag{2·13}$$

The first term in brackets in (2·13) is asymptotically distributed as M, and by (2·4) the second term can be written as

$$\max\left(0, -y_r, \ldots, -\sum_{i=1}^{r} y_i, -\sum_{i=1}^{r} y_i+y_1', \ldots, -\sum_{i=1}^{r} y_i+\sum_{j=1}^{n} y_j', \ldots\right). \tag{2·14}$$

Recalling that y_i is a $N(-\delta,1)$ random variable and y_j' is a $N(-2\Delta+\delta,1)$ random variable, it can be deduced from (2·13) and (2·14) that

$$\mathrm{pr}(\Lambda_2^* \leqslant x|\tau = \tau_0+r;\delta,\Delta) = \alpha(x;\delta)p(x,r,2\Delta;2\Delta-\delta), \tag{2·15}$$

where

$$p(x,r,\lambda;\mu) = \mathrm{pr}\left\{\sum_{i=1}^{n} z_i \leqslant x-\lambda\min(r,n), n\geqslant 1 \middle| z_i \text{ is } N(-\mu,1) \text{ random variable}\right\}.$$

Similarly it can be shown that

$$\mathrm{pr}(\Lambda_2^* \leqslant x|\tau = \tau_0-r;\delta,\Delta) = \alpha(x;2\Delta-\delta)p(x,r,2\Delta;\delta).$$

The probability $p(x,r,\lambda;\mu)$ is an awkward absorption probability which we shall not discuss here. However, to illustrate the power of Λ_2^* it will suffice to consider $r = +1, +2$, in which case it is convenient to use the integral representation

$$p(x,r,\lambda;\mu) = \int_{-\infty}^{x}\ldots\int_{-\infty}^{x}\alpha(x-u_r;\mu)\prod_{i=1}^{r}\phi(u_i-u_{i-1}-\lambda+\mu)\,du_i \quad (u_0 = 0). \tag{2·16}$$

Note that $p(x,1,\lambda;\mu) = \alpha(x-\lambda;\mu)$; see (2·8).

Using numerical integration of (2·16), we have computed values of the power of Λ_2^* from (2·15) for a few values of δ and Δ. To illustrate the results, Table 4 gives respective powers of the 98% two-sided tests using $\tilde{\tau}$, $\hat{\tau}$, Λ_2^* and Λ_2 when $\tau-\tau_0 = 0, \pm 1, \pm 2$ in the cases $\delta = 0{\cdot}5$, $0{\cdot}7$ and $1{\cdot}0$ with $\Delta = 1{\cdot}0$. In practice Δ would be estimated, but for the sake of comparison we assume Δ to be correctly estimated as $1{\cdot}0$. The superiority of Λ_2^* over $\tilde{\tau}$

is considerable, as one might expect. Even with $\delta = 0{\cdot}5$, Λ_2^* is more powerful than the maximum likelihood estimate $\hat{\tau}$. Also considerable power is lost on one side in using Λ_2^* rather than Λ_2 when δ is not close to Δ. Note that $\hat{\tau}$ is not uniformly better than $\tilde{\tau}$, since $\mathrm{pr}\,(\tilde{\tau} > \tau + a)$ decreases as δ decreases for any a; see Fig. 2.

In practice the parameter Δ will be unknown, and is therefore a nuisance parameter in the inference procedures described above. The estimate

$$\tilde{\Delta} = \frac{S_{\tilde{\tau}} - S_T}{2\sigma(T - \tilde{\tau})} + \tfrac{1}{2}\delta \tag{2·17}$$

Table 4. *Asymptotic powers of* 98% *two-sided tests of* H_0: $\tau = \tau_0$ *when* $\Delta = 1{\cdot}0$

δ	test statistic \ $\tau - \tau_0$	-2	-1	0	1	2
0·5	$\tilde{\tau}$	0·026	0·022	0·020	0·024	0·061
	Λ_2^*	0·531	0·148	0·020	0·023	0·073
0·7	$\tilde{\tau}$	0·029	0·021	0·020	0·031	0·095
	Λ_2^*	0·668	0·269	0·020	0·094	0·316
1·0	$\tilde{\tau}, \hat{\tau}$	0·055	0·026	0·020	0·026	0·055
	Λ_2^*, Λ_2	0·621	0·282	0·020	0·282	0·621

is consistent and asymptotically normal as τ and $T - \tau \to \infty$, so that consistent estimation of percentage points for Λ_2^*, or $\tilde{\tau}$, is possible. The maximum likelihood estimate of Δ is (Hinkley, 1970)

$$\hat{\Delta} = \frac{Z_{\hat{\tau}}}{2\surd(T - \hat{\tau})}. \tag{2·18}$$

It should be emphasized that $\hat{\Delta}$, and of course $\tilde{\Delta}$, are not sufficient for Δ: the minimal sufficient set for τ and Δ is the original sample.

Thus far we have seen that the likelihood ratio statistic is more powerful than the cusum statistic Λ_2^* when $\delta \neq \Delta$. Unfortunately we cannot choose $\delta = \Delta$, so optimum large-sample inference would appear to preclude a cusum statistic. But we can compute $\tilde{\Delta}$, and re-compute the sequence $S_1, \ldots, S_T$ using $\tilde{\Delta}$ in place of δ. Let these new sums be denoted by

$$\tilde{S}_t = \sum_{i=1}^{t} (x_i - \theta_0 + \tilde{\Delta}\sigma). \tag{2·19}$$

Then we have a new cusum statistic

$$\tilde{\Lambda}_2 = \max_t\,(\tilde{S}_t - \tilde{S}_{\tau_0}). \tag{2·20}$$

Conditional on $\tilde{\Delta}$, $\tilde{\Lambda}_2$ has asymptotic distribution $P_2^*(x; \tilde{\Delta}, \tilde{\Delta})$. Therefore

$$\mathrm{pr}\,(\tilde{\Lambda}_2 \geqslant l_\alpha(\tilde{\Delta}, \tilde{\Delta})) \sim \alpha, \tag{2·21}$$

where $l_\alpha(\Delta, \Delta)$ is defined in (2·12). The consistency of $\tilde{\Delta}$ implies that $\tilde{\Lambda}_2$ and Λ_2 are asymptotically equivalent, and hence that the test based on (2·21) is asymptotically efficient. Note that $\tilde{\Delta}$ is being used as a sufficient statistic: since $\tilde{\tau} - \tau$ is $O_p(1)$, $\tilde{\Delta}$ is asymptotically sufficient.

2·4. *Both means unknown*

If the initial mean θ_0 is unknown, it is still possible to use the cusum procedure with, conditional on T,

$$S_t = \sum_{i=1}^{t} (x_i - \bar{x}_T), \quad \bar{x}_T = T^{-1} \sum_{j=1}^{T} x_j.$$

The test rule (1·2) still applies, as do the definitions of $\hat{\tau}$ and Λ_2^*. Average run length properties of the scheme can now be controlled only by h; δ is replaced by τ/T, which cannot be controlled. The asymptotic theory of §§ 2·2 and 2·3 goes through, the argument resembling that of Hinkley (1970, § 3·2).

The asymptotically efficient cusum statistic $\tilde{\Lambda}_2$ of § 2·3 is now derived using the consistent cusum estimates $\tilde{\theta}_0$ and $\tilde{\theta}_1$ given by

$$\tilde{\theta}_0 = \hat{\tau}^{-1} \sum_{i=1}^{\hat{\tau}} x_i, \quad \tilde{\theta}_1 = (T-\hat{\tau})^{-1} \sum_{i=\hat{\tau}+1}^{T} x_i.$$

Then the new cumulative sums are

$$\tilde{S}_t = \sum_{i=1}^{t} (x_i - \tilde{\theta}_0 + \tilde{\Delta}\sigma) \quad (t = 1, \ldots, T)$$

with $\tilde{\Delta}\sigma = \frac{1}{2}(\tilde{\theta}_0 - \tilde{\theta}_1)$. The statistic $\tilde{\Lambda}_2$ is defined by (2·20).

3. SEQUENTIAL CUSUM SCHEMES

In § 2 we concentrated entirely on large-sample properties of the cusum change-point statistics $\hat{\tau}$, Λ_2^*, $\hat{\tau}$ and Λ_2 assuming $T-\tau$ to be large. The results are therefore strictly valid only when T is a large fixed sample size, or when h is very large. The qualitative comparison of the inference methods in § 2·3 will be valid generally, but will the asymptotic distributions of $\hat{\tau}$, Λ_2^*, $\hat{\tau}$, Λ_2 and $\tilde{\Lambda}_2$ be good approximations for the general sequential scheme? This important question is the subject of this section. First, we make some general theoretical remarks, and then some Monte Carlo results are described.

3·1. *General remarks*

It should be pointed out that for fixed sample size T, the asymptotic distributions of $\hat{\tau}$ and Λ_2^* are good approximations when τ and $T-\tau$ are moderately large. This can be deduced from the random walk representation of $\{S_t - S_\tau\}$ in § 2·1. Some empirical evidence was given by Hinkley (1970), the conclusion being that it was necessary for, say, 0·999 of the asymptotic distribution of $\hat{\tau}-\tau$ to be within $(-\tau, T-\tau)$. Therefore the suitability of asymptotic theory for sequential sampling will depend mostly on the effect of T not getting much past τ.

Suppose N is a fixed integer so large that $\mathrm{pr}(T > \tau + N)$ is negligible, and consider all sequences $(x_1, \ldots, x_{\tau+N})$ generated by the model (2·1). Let $\hat{\tau}_S$ be the cusum estimate of τ for sequential sampling, when T is determined by the stopping rule (1·2). Also let $\hat{\tau}_F$ be the corresponding estimate for the fixed sample size $\tau + N$. Now define

$$M_t = \max_{i \leqslant t} S_i,$$

with S_i given by (1·1). Since $T \leqslant \tau + N$ almost always, $\hat{\tau}_S \leqslant \hat{\tau}_F$ almost always. If τ is reasonably large, the distributions of $\hat{\tau}_F$ and related statistics will be close to the asymptotic distributions derived in § 2.

The difference between the distributions of $\hat{\tau}_S$ and $\hat{\tau}_F$ depends on how often M_T and $M_{\tau+N}$ differ, and on how often T is less than τ. For if

$$p_{1,t} = \mathrm{pr}(T < t), \quad p_2 = \mathrm{pr}(M_T < M_{\tau+N} \mid T \geqslant \tau),$$

then

$$\text{pr}(\tilde{\tau}_S < \tilde{\tau}_F) < p_{1,\tau} + p_2. \tag{3·1}$$

Typically p_2 will be small: since $M_T - S_T > h$, $p_2 < 1 - \alpha(h; 2\Delta - \delta)$. The probability $p_{1,\tau}$, on the other hand, can be nonnegligible. An approximation to $p_{1,t}$ is (van Dobben de Bruyn, 1969, p. 71)

$$p_{1,t} \simeq 1 - e^{-t/L},$$

where $L = \lim_{\tau \to \infty} E(T)$ is the average run length at mean θ_0. As an example, when $\tau = 50$, $\delta = \Delta = 0·6$ and $h = 4·0$ we find $p_2 < 0·004$ and $p_{1,t} \simeq 0·072$. The upper bound (3·1) is quite loose, but in this example $p_{1,\frac{1}{2}\tau} \simeq 0·036$ and $\text{pr}(\tilde{\tau}_F \leqslant \frac{1}{2}\tau) \simeq 0·001$, so that the distribution functions of $\tilde{\tau}_F$ and $\tilde{\tau}_S$ differ by as much as 0·035 in the lower tails. As L increases and τ decreases, the two distributions get closer; if $\tau = 25$, $\delta = \Delta = 1·0$ and $h = 3·0$, then $p_2 < 0·001$ and $p_{1,\tau} \simeq 0·013$.

A more sophisticated approach would lead to a better comparison of $\tilde{\tau}_F$ and $\tilde{\tau}_S$, but the above discussion gives an outline of the effect of sequential sampling on the estimation of τ.

For inference purposes it is more important to consider the effect on Λ_2^*. Recall that Λ_2^* is defined by

$$\Lambda_2^* = \frac{1}{\sigma}(S_{\tilde{\tau}} - S_{\tau_0}),$$

and let $\Lambda_{2,S}^*$ and $\Lambda_{2,F}^*$ correspond to $\tilde{\tau}_S$ and $\tilde{\tau}_F$. Now there is a difficulty with the interpretation of $\Lambda_{2,S}^*$, since if T is less than τ_0, then S_{τ_0} is not automatically computed. But suppose that S_{τ_0} is computed. Then $S_{\tilde{\tau}_S} \leqslant S_{\tilde{\tau}_F}$, from which it follows that

$$\text{pr}(\Lambda_{2,S}^* > x) \leqslant \text{pr}(\Lambda_{2,F}^* > x).$$

This means that percentage points of the asymptotic distribution of Λ_2^* are conservative for testing in the sequential case. In fact it is not too difficult to see that

$$\text{pr}(\Lambda_{2,F}^* > x) - \text{pr}(\Lambda_{2,S}^* > x) \simeq (p_2 + p_{1,\tau_0})\,\text{pr}(\Lambda_F^* > x), \tag{3·2}$$

which is relatively small.

Now suppose that sampling does not continue beyond the Tth observation even if $T < \tau_0$. Then a revised test of H_0: $\tau = \tau_0$ is required. A natural test is

$$\text{reject } H_0 \text{ if } T < \tau_0 \text{ or if } \Lambda_{2,S}^* \text{ is significant.} \tag{3·3}$$

The size of this test is not more than

$$p_{1,\tau_0} + (1 - p_{1,\tau_0})\beta \tag{3·4}$$

if the upper 100β percentile of Λ_2^* is used. Typically the size will be appreciably more than β; see our previous discussion of $\tilde{\tau}_S$. Naturally power is lost if $\Lambda_{2,S}^*$ is not always used, since this effectively means not examining the likelihood of the parameter value, τ_0, under test. Nonetheless it might be useful to study the revised test (3·3) in detail.

Remarks similar to those above apply to the one-sided test statistic Λ_1^* in (2·11).

When T is variable, the maximum likelihood estimate $\hat{\tau}$ and the likelihood ratio statistic Λ_2 are still as defined in § 2, but the relationship (2·3) between Z_t and S_t prohibits simple analysis. Moreover, the asymptotic distributions of $\hat{\tau}$ and Λ_2 rely on the sequence $\{Z_t - Z_\tau\}$ being asymptotically equivalent to a pair of random walks such as **W** and **W**′ in (2·5). In the sequential case with $T - \tau$ small, this equivalence will not be a reasonable approximation. Therefore the asymptotic properties of $\hat{\tau}$ and Λ_2 will be worse approximations than those of $\tilde{\tau}$ and Λ_2^*. Some empirical results are given in § 3·2.

Even if the asymptotic distributions of $\tilde{\tau}$, Λ_2^*, $\hat{\tau}$ and Λ_2 are excellent approximations for the sequential case, the dependence of those distributions on Δ is a source of practical difficulty. The two estimates $\tilde{\Delta}$ and $\hat{\Delta}$ in (2·17) and (2·18) are consistent and asymptotically normal, but when $T-\tau$ is small we might expect large variances for both estimates, and possibly nonnegligible biases. We have derived no satisfactory theoretical analysis, although some properties of $\tilde{\Delta}$ are obvious. First, as $2\Delta-\delta$ decreases the bias of $\tilde{\Delta}$ increases, since by (2·17) and (1·2)

$$\tilde{\Delta} > \tfrac{1}{2}\delta + \frac{h}{T-\tilde{\tau}}. \tag{3·5}$$

The maximizing properties of $\tilde{\tau}$ and $\hat{\tau}$ also indicate a positive bias in $\tilde{\Delta}$ and $\hat{\Delta}$. Secondly, the asymptotic variance

$$\sigma_{\Delta}^2 = \frac{1}{4(T-\tau)} \tag{3·6}$$

of $\tilde{\Delta}$ and $\hat{\Delta}$ will underestimate the small-sample variance because of the nonnegligible variability of $T-\tilde{\tau}$ and $T-\hat{\tau}$ when $T-\tau$ is small.

To use the test statistic Λ_2^*, we have to estimate the appropriate significance point $l_\alpha(\delta,\Delta)$ using $\tilde{\Delta}$. The first-order estimate

$$\tilde{l}_\alpha(\delta,\Delta) = l_\alpha(\delta,\tilde{\Delta}) \tag{3·7}$$

will tend to have a positive bias, since

$$E\{l_\alpha(\delta,\tilde{\Delta})\} \simeq l_\alpha(\delta,\Delta) + \tfrac{1}{2}\sigma_{\Delta}^2 \frac{d^2 l_\alpha(\delta,\tilde{\Delta})}{d\Delta^2} + o\left(\frac{1}{T-\tilde{\tau}}\right).$$

The $O\{1/(T-\tau)\}$ term in the bias is removed by using the second-order estimate

$$\tilde{\tilde{l}}_\alpha(\delta,\Delta) = l_\alpha(\delta,\tilde{\Delta}) - \frac{1}{8(T-\tilde{\tau})}\frac{d^2 l_\alpha(\delta,\tilde{\Delta})}{d\Delta^2}, \tag{3·8}$$

σ_{Δ}^2 being replaced by its estimate $1/\{4(T-\tilde{\tau})\}$. The second derivative of $l_\alpha(\delta,\Delta)$ can be crudely approximated by differencing the entries in Table 3. For $\Delta > \delta$ this procedure should provide a good estimate of $l_\alpha(\delta,\Delta)$. Some empirical results using (3·7) and (3·8) are given in § 3·2.

The conditional test (2·21) using $\tilde{\Lambda}_2$ does not depend on an unknown significance point. The distribution of $\tilde{\Lambda}_2$ will converge more slowly than that of Λ_2^* but since $\{S_t - S_\tau\}$ does define two random walks, unlike $\{Z_t - Z_\tau\}$, one would expect the distribution of $\tilde{\Lambda}_2$ to converge more quickly than that of Λ_2. Note that the strong dependence on $\tilde{\Delta}$ of the $\tilde{\Lambda}_2$ test will probably make a significant bias in $\tilde{\Delta}$ more disturbing than it would be for the Λ_2^* and Λ_2 tests. Unfortunately the relevant small-sample theory is difficult, and no quantitative results have been derived. Some empirical results are given in § 3·2.

3·2. *Monte Carlo results*

In the previous section several comments were made about the distributions of $\tilde{\tau}$, Λ_2^*, $\tilde{\Delta}$ and their likelihood counterparts in the sequential cusum scheme. Also some remarks were made concerning practical inference about τ. In order to get a better idea of what happens, cusum schemes were simulated for several sets of parameters. We generated 1000 sequences of 100 pseudo-random normal deviates, and used these sequences for every case. Results are described here for five cases, which are listed in Table 5 together with their average run lengths at means θ_0 and θ_1.

The more important results are those for the test statistics Λ_2^*, Λ_2 and $\tilde{\Lambda}_2$. Table 6 gives the empirical sizes of the 95% tests, together with the powers at $\tau = \tau_0 + 1$ of Λ_2^* and $\tilde{\Lambda}_2$. The results confirm that the asymptotic distribution of Λ_2^* is an adequate approximation, while that of Λ_2 is not. Use of the second-order estimate $\tilde{\tilde{l}}_{0\cdot05}(\delta, \Delta)$ for the Λ_2^* test gives very good agreement with theoretical size except in cases (i) and (ii), where $\tilde{\Delta}$ has a large positive

Table 5. *Cases for which cusum scheme was simulated*

Case	δ	Δ	τ	h	Average run length at θ_0	Average run length at θ_1
(i)	0·5	0·5	25	4·0	340	8·4
(ii)	0·5	0·5	50	4·0	340	8·4
(iii)	0·5	1·0	25	4·0	340	3·3
(iv)	0·7	1·0	25	4·0	1300	3·8
(v)	0·7	1·0	25	3·0	330	3·0

Table 6. *Empirical sizes of Λ_2^*, Λ_2 and $\tilde{\Lambda}_2$ tests for cases in Table 5, with power at $\tau = \tau_0 + 1$ for Λ_2^* and $\tilde{\Lambda}_2$*

Test	(i)	(ii)	(iii)	(iv)	(v)	
$\Lambda_2^* \geq l_{0\cdot05}(\delta, \Delta)$	0·048	0·044	0·041	0·043	0·043	Size
	0·083	0·086	0·061	0·202	0·193	Power at $\tau = \tau_0 + 1$
$\Lambda_2^* \geq \tilde{l}_{0\cdot05}(\delta, \Delta)$	0·062	0·062	0·031	0·030	0·031	Size
$\Lambda_2^* \geq \tilde{\tilde{l}}_{0\cdot05}(\delta, \Delta)$	0·071	0·076	0·040	0·046	0·050	Size
$\Lambda_2 \geq l_{0\cdot05}(\Delta, \Delta)$	0·057	0·065	0·034	0·035	0·035	Size
$\tilde{\Lambda}_2 \geq l_{0\cdot05}(\tilde{\Delta}, \tilde{\Delta})$	0·132	0·125	0·045	0·063	0·060	Size
	0·269	0·268	0·398	0·484	0·475	Power at $\tau = \tau_0 + 1$

Table 7. *Empirical moments of $\tilde{\Delta}$ and $\hat{\Delta}$*

Case	Δ	ave $(\tilde{\Delta})$	var $(\tilde{\Delta})$	ave $\left(\dfrac{0\cdot25}{T-\tilde{\tau}}\right)$	ave $(\hat{\Delta})$	var $(\hat{\Delta})$
(i)	0·5	0·67	0·046	0·045	0·77	0·086
(ii)	0·5	0·66	0·038	0·042	0·76	0·082
(iii)	1·0	0·94	0·093	0·068	1·12	0·129
(iv)	1·0	1·05	0·074	0·070	1·14	0·103
(v)	1·0	1·05	0·096	0·090	1·15	0·139

bias (Table 7). The $\tilde{\Lambda}_2$ test has approximately correct size, again with the exception of cases (i) and (ii). The powers of Λ_2^* and $\tilde{\Lambda}_2$ correspond quite well to the asymptotic values; for example, in case (iv) the asymptotic powers are respectively 0·235 and 0·452.

The empirical distributions of $\tilde{\Delta}$ and $\hat{\Delta}$ deviated sharply from normality. The means and variances are given in Table 7, together with the average estimated variance $0\cdot25\,(T-\tilde{\tau})^{-1}$ of $\tilde{\Delta}$. Both $\tilde{\Delta}$ and $\hat{\Delta}$ have large biases in cases (i) and (ii), reflecting the fact that the average of the lower bound (3·5) is greater than Δ. In the other three cases $\tilde{\Delta}$ has negligible bias, but $\hat{\Delta}$ is still strongly biased. Note that $\tilde{\Delta}$ always has smaller variance than $\hat{\Delta}$, making $\tilde{\Delta}$ the preferable estimate. The average estimate of var $(\tilde{\Delta})$ is surprisingly good.

The empirical distributions of $\tilde{\tau}_S$ corroborated the remarks in § 3·1, there being significantly large probabilities in the lower tails due to occurrence of $T < \tau$. The central frequencies were quite close to the asymptotic probabilities. Table 8 gives the biases of $\tilde{\tau}$ and $\hat{\tau}$, also the asymptotic biases from Table 1. In addition the table gives the biases of the estimate derived from the sequence $\{\tilde{S}_t\}$, which is denoted by $\tilde{\tau}(\tilde{\Delta})$. This last estimate is less biased than $\tilde{\tau}$, but $\hat{\tau}$ has the least bias. The empirical distribution of $\hat{\tau}$ had a longer lower tail than the asymptotic distribution, as did the distribution of $\tilde{\tau}(\tilde{\Delta})$.

Table 8. *Empirical biases of $\hat{\tau}$, $\tilde{\tau}(\tilde{\Delta})$ and $\tilde{\tau}$*

Case	Bias of $\hat{\tau}$	Bias of $\tilde{\tau}(\tilde{\Delta})$	Bias of $\tilde{\tau}$	Asymptotic bias of $\tilde{\tau}$
(i)	$-0{\cdot}58$	$-0{\cdot}89$	$-0{\cdot}96$	$0{\cdot}0$
(ii)	$-3{\cdot}77$	$-3{\cdot}76$	$-3{\cdot}78$	$0{\cdot}0$
(iii)	$-1{\cdot}46$	$-2{\cdot}06$	$-2{\cdot}47$	$-1{\cdot}69$
(iv)	$-0{\cdot}60$	$-0{\cdot}75$	$-0{\cdot}92$	$-0{\cdot}68$
(v)	$-1{\cdot}41$	$-1{\cdot}47$	$-1{\cdot}61$	$-0{\cdot}68$

4. DISCUSSION

The theoretical and empirical studies in §§2 and 3 indicate that $\tilde{\tau}$ is not as good an estimate as $\hat{\tau}$, although the latter is more difficult to compute. The revised cusum estimate $\tilde{\tau}(\tilde{\Delta})$ is better than $\tilde{\tau}$ and can be computed by a simple rotation of $\{S_t\}$ to $\{\tilde{S}_t\}$. All three estimates are very inefficient test statistics (Table 4). Of the three likelihood-type statistics, $\tilde{\Lambda}_2$ is preferable since it is easier to use than either Λ_2^* or Λ_2 and is approximately as powerful as Λ_2. The asymptotic distributions of these statistics are good approximations provided that the cusums are computed up to the index under test. This remark is particularly relevant to the construction of confidence limits on τ, because then all values of τ_0 are open to test, so that derivation of an upper confidence limit involves computation of an unknown number of cusums beyond S_T. In practice it would often make sense to have a lower confidence limit only, in which case the statistic Λ_1^*, or the more powerful analogue $\tilde{\Lambda}_1$ from $\{\tilde{S}_t\}$, would be appropriate.

It should be emphasized that if $\Delta \simeq \delta$, then the estimate $\tilde{\Delta}$ can have a large bias, making the asymptotic properties of tests poor approximations. Further work on properties of Λ_2^*, $\tilde{\tau}$, $\tilde{\Delta}$ and other statistics in the sequential case would therefore be of value.

Inference about change-points using the cumulative sum test scheme is a valuable tool in situations where a sequence $x_1, \ldots, x_T$ contains more than one change-point. The Bayesian analysis by Chernoff & Zacks (1964) is also relevant here.

Robustness of the inference procedures has not been discussed in this paper. Theoretical analysis of the random walks (2·5) is not difficult for nonnormal distributions, so robustness against nonnormality could be determined quite easily. Note that $\tilde{\Lambda}_2$ and the likelihood ratio test statistic are only asymptotically equivalent in the normal case, so that the efficiency of $\tilde{\Lambda}_2$ for nonnormal cases is also of interest. More difficult is the question of robustness against correlation between variables. Some unpublished empirical results by Goldsmith indicate lack of robustness in this situation. When the change in mean from θ_0 is to a trend rather than a constant, the random walk $\{S_t - S_\tau, t \geqslant \tau\}$ no longer has independently and identically distributed increments, which complicates theoretical analysis. However, the properties of Λ_2^* and other statistics are probably quite robust against slow trends.

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